

The spectral sequence of an extraordinary cohomology theory

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1. *Introduction.* Given an 'extraordinary cohomology theory', that is, a cohomology theory satisfying all the axioms of Eilenberg and Steenrod (7) except the dimension axiom, it is well known that there exists a spectral sequence relating the ordinary cohomology of a space with the extraordinary theory (see, for example, (3) in the case of $K^*(X)$). Obviously, it would be useful to know the differentials in this spectral sequence, and it is the purpose of this paper to identify them in terms of cohomology operations defined by certain k -invariants. We shall make use of E. H. Brown's recent theorem (5) on the representability of extraordinary cohomology theories, to construct a second spectral sequence, in which the differentials are readily identifiable, which we shall prove is isomorphic to the usual one.

The plan of the paper is as follows: in § 2 we summarize the necessary information about extraordinary cohomology theories, killing homotopy groups, and k -invariants. § 3 contains the statements of our main results, which are proved in § 4; we end in § 5 with a few examples and applications.

2. *Preliminaries.* 2.1. *Extraordinary cohomology theories.* (For details see (5).) Let $\bar{H}^p(X, A)$ be an extraordinary cohomology theory, satisfying all the Eilenberg-Steenrod axioms (7) except the dimension axiom, which is replaced by the condition that the groups $\bar{H}^p(P)$ are countable, where P is a single point. Such a cohomology theory is *representable*, in the following sense: we define an Ω -spectrum Y to be a set of spaces $\{Y_q\}$ with base points, together with homotopy equivalences $h_q: Y_q \rightarrow \Omega_{q+1} Y_q$, where $\Omega_{q+1} Y_q$ is the space of loops based at the base point, and h_q takes the base point into the constant loop. Then Brown's theorem states that, given a cohomology theory $\bar{H}^p(X, A)$ as above, there exists an Ω -spectrum Y , such that

$$\bar{H}^p(X, A) = [X/A, Y_p]_0,$$

where $[A, B]_0$ denotes the set of homotopy classes of maps from A to B which preserve base points, and X/A denotes the space with base point p , formed from

$$X \cup (A \times I) \cup \{p\},$$

where $p \notin X$, by identifying a and $(a, 1)$ for all $a \in A$ and $A \times \{0\}$ with p . Thus

$$\begin{aligned} \bar{H}^p(X) &= \bar{H}^p(X, \phi) \quad (\text{by definition}) \\ &= [X \cup \{p\}, Y_p]_0 \\ &= [X, Y_p], \end{aligned}$$

the set of homotopy classes of maps from X to Y_p (ignoring base points).

We may identify the maps i^* , j^* and δ^* of the exact cohomology sequence as follows: using Puppe's sequence of spaces and maps (see (13))

$$A \xrightarrow{i} X \xrightarrow{Pi} X/A \xrightarrow{Qi} SA \rightarrow SX \rightarrow S(X/A) \rightarrow \dots$$

we define

$$i^*: \bar{H}^p(X) \rightarrow \bar{H}^p(A) \quad \text{and} \quad j^*: \bar{H}^p(X, A) \rightarrow \bar{H}^p(X)$$

to be the maps induced by i and Pi , respectively, while $\delta^*: \bar{H}^p(A) \rightarrow \bar{H}^{p+1}(X, A)$ is defined to be the composite

$$[A^+, Y_p]_0 \xrightarrow{h_p} [A^+, \Omega Y_{q+1}]_0 = [SA^+, Y_{q+1}]_0 \xrightarrow{(Qi)^*} [X/A, Y_{q+1}]_0,$$

where $A^+ = A \cup \{p\}$, and $(Qi)^*$ is induced by Qi . The sequence

$$\dots \rightarrow \bar{H}^{p-1}(A) \xrightarrow{\delta^*} \bar{H}^p(X, A) \xrightarrow{j^*} \bar{H}^p(X) \xrightarrow{i^*} \bar{H}^p(A) \rightarrow \dots$$

is then exact, and coincides with the usual exact sequence.

2.2. Killing homotopy groups. We use the notation of (1), with a slight modification. Let X and Y be connected CW-complexes. Then Y is a *space of type* $X(1, n)$ if

(i) There exists a map $f: X \rightarrow Y$, such that $f_*: \pi_r(X) \rightarrow \pi_r(Y)$ is an isomorphism, $1 \leq r \leq n$.

(ii) $\pi_r(Y) = 0$, ($r > n$).

If Y is such a space, of type $X(1, n)$ and $m \leq n$, we define a *space of type* $X(m, n)$ to be a space of type $Y(m, \infty)$, with the obvious definition of $Y(m, \infty)$. Then $X(m, n)$ is unique, up to homotopy equivalence, and, given $r \leq s < t$, there is a Serre fibring

$$X(s+1, t) \rightarrow X(r, t) \rightarrow X(r, s).$$

According to Peterson (11), given a Serre fibring $F \xrightarrow{j} E \xrightarrow{p} B$, there exists a map $i: \Omega B \rightarrow F$ such that the induced sequence

$$[X, B] \xrightarrow{p^*} [X, E] \xrightarrow{j^*} [X, F] \xrightarrow{i^*} [X, \Omega B] \xrightarrow{(\Omega p)^*} [X, \Omega E] \rightarrow \dots$$

is exact; this may be expressed by saying that

$$\dots \rightarrow \Omega E \xrightarrow{\Omega p} \Omega B \xrightarrow{i} F \xrightarrow{j} E \xrightarrow{p} B$$

is an *exact sequence of spaces* (respecting base points, in fact). Applying this to an Ω -spectrum $\mathbf{Y}$, we obtain (for $r \leq s < t$) an exact sequence of spaces

$$\dots \rightarrow Y_q(r, t) \rightarrow Y_q(r, s) \xrightarrow{\delta} Y_{q+1}(s+2, t+1) \rightarrow Y_{q+1}(r+1, t+1) \rightarrow \dots,$$

where δ is the composite

$$Y_q(r, s) \xrightarrow{h_q} \Omega[Y_{q+1}(r+1, s+1)] \xrightarrow{i} Y_{q+1}(s+2, t+1).$$

2.3. k -invariants and cohomology operations. If in the above sequence $s+1 = t$, we obtain the sequence

$$\dots \rightarrow Y_q(r, s+1) \rightarrow Y_q(r, s) \xrightarrow{\delta} Y_{q+1}(s+2) \rightarrow Y_{q+1}(r+1, s+2) \rightarrow \dots,$$

where $Y_{q+1}(s+2)$ denotes

$$\begin{aligned} Y_{q+1}(s+2, s+2) &= K(\pi_{s+2}(Y_{q+1}), s+2) \\ &= K(\pi_{s+1}(Y_q), s+2). \end{aligned}$$

Thus the homotopy class of δ represents an element of

$$H^{s+2}(Y_q(r, s); \pi_{s+1}(Y_q)),$$

viz. the k -invariant $k^{r, s+1}$ of Y_q . These k -invariants may be used to define (higher order) cohomology operations in the following way (cf. (12)).

An element of $H^r(X; \pi_r(Y_q))$ may be regarded as a homotopy class $[f]$ of maps $f: X \rightarrow Y_q(r)$. Suppose the composite

$$X \xrightarrow{f} Y_q(r) \rightarrow Y_{q+1}(r+2, s+1)$$

is homotopic to zero. Then there exists a map $g: X \rightarrow Y_q(r, s)$, such that the following diagram is homotopy-commutative.

$$\begin{array}{ccc} & Y_q(r, s) & \\ g \nearrow & \downarrow \pi & \\ X & \xrightarrow{f} & Y_q(r) \end{array}$$

Then the composite $g\delta: X \rightarrow Y_{q+1}(s+2)$ represents an element of

$$H^{s+2}(X; \pi_{s+1}(Y_q)).$$

(Here, as always, $g\delta$ means 'first g , then δ '.) Of course, g is not in general unique, but we may define the cohomology operation $[k^{r, s+1}]$ by

$$[k^{r, s+1}][f] = \{[g\delta]\}$$

for those maps g such that $g\pi \simeq f$. In fact this defines a homomorphism from the subgroup of $H^r(X; \pi_r(Y_q))$ on which $[k^{r, s}]$ is defined and includes zero, to a certain quotient group of $H^{s+2}(X; \pi_{s+1}(Y_q))$ (see (12)).

3. *Results.* In the sequel, let X be a finite simplicial complex (our results will hold also for finite CW-complexes, since by a theorem of Whitehead (14), any finite CW-complex is of the homotopy type of a finite simplicial complex). Denote by X^q the q -skeleton of X ; by filtering over skeletons of X , we obtain

THEOREM 3.1. *There exists a spectral sequence $E_r^{p, q}$ ($r \geq 1$, $p \geq 0$) with differentials $d_r: E_r^{p, q} \rightarrow E_r^{p+r, q-r+1}$, such that*

(a) $E_1^{p, q} = C^p(X; \pi_p(Y_{p+q}))$, d_1 being the ordinary co-boundary operator;

(b) $E_2^{p, q} = H^p(X; \pi_p(Y_{p+q}))$;

(c) $E_\infty^{p, q} = G_p \bar{H}^{p+q}(X) = \bar{H}_p^{p+q} / \bar{H}_{p+1}^{p+q}(X)$, where $\bar{H}_p^n(X) = \text{Ker}[\bar{H}^n(X) \rightarrow \bar{H}^n(X^{p-1})]$.

(Compare the theorem of Atiyah and Hirzebruch on $K^*(X)$ (3).) Note that

$$\pi_p(Y_{p+q}) \simeq \pi_0(Y_q) = [S^0, Y_q]_0 = \bar{H}^q(P),$$

where P is a single point. Alternatively, by filtering over the Postnikov systems of the spaces Y_q , we obtain

THEOREM 3.2. *There exists a spectral sequence $\bar{E}_r^{p, q}$ ($r \geq 2$, $p \geq 0$) with differentials $\bar{d}_r: \bar{E}_r^{p, q} \rightarrow \bar{E}_r^{p+r, q-r+1}$, such that*

(a) $\bar{E}_2^{p, q} = \bar{H}^p(X; \pi_p(Y_{p+q}))$;

(b) $\bar{E}_\infty^{p, q} = \bar{G}_p \bar{H}^{p+q}(X) = {}_p \bar{H}^{p+q}(X) / {}_{p+1} \bar{H}^{p+q}(X)$, where

$${}_p \bar{H}^n(X) = \text{Ker}[\bar{H}^n(X) \rightarrow [X, Y_n(0, p-1)]].$$

The resemblance between these two spectral sequences is no coincidence; in fact we shall prove

THEOREM 3.3. *There exist isomorphisms $\phi_r^{p,q}: E_r^{p,q} \rightarrow \bar{E}_r^{p,q}$ ($r \geq 2$), such that*

$$\phi_r^{p,q} d_r = d_r \phi_r^{p+q, q-r+1}.$$

From this it follows that we may identify the two spectral sequences (at least for $r \geq 2$); in particular $G_p \bar{H}^n(X) = \bar{G}_p \bar{H}^n(X)$, and the differentials in the two spectral sequences are the same. The differentials $\bar{d}_r$ are readily identifiable; in fact we have

THEOREM 3.4. *$\bar{d}_r: \bar{E}_r^{p,q} \rightarrow \bar{E}_r^{p+r, q-r+1}$ is induced by the cohomology operation defined by the k -invariant $k^{p, p+r-2} \in H^{p+r}(Y_{p+q}(p, p+r-2); \pi_{p+r-1}(Y_{p+q}))$.*

In particular, the differentials in the spectral sequence for $K^*(X)(3)$ are given by the k -invariants of BU ; we return to this point in § 5.

4. *Proofs.* 4.1. *Proof of Theorem 3.1.* We shall define our spectral sequences by means of exact couples, referring the reader to (9) for details. For Theorem 3.1, define an exact couple (A, C) as follows:

$$\text{let } C_{p,q} = \bar{H}^{p+q}(X^p, X^{p-1}),$$

$$A_{p,q} = \bar{H}^{p+q}(X^p),$$

where the maps

$$f_{p,q}: A_{p,q} \rightarrow A_{p-1, q+1},$$

$$g_{p,q}: A_{p,q} \rightarrow C_{p+1, q},$$

and

$$h_{p,q}: C_{p,q} \rightarrow A_{p,q}$$

are defined by i^* , j^* , and δ^* , respectively.

Here,

$$\begin{aligned} E_1^{p,q} &= C_{p,q} = \bar{H}^{p+q}(X^p, X^{p-1}) \\ &= \sum_i \bar{H}^{p+q}(\sigma_i^p, \dot{\sigma}_i^p), \end{aligned}$$

where σ_i^p runs through all p -simplexes of X . But $\sigma^p/\dot{\sigma}^p = S^p$, and so

$$\bar{H}^{p+q}(\sigma^p, \dot{\sigma}^p) = [S^p, Y_{p+q}]_0 = \pi_p(Y_{p+q}).$$

Hence $E_1^{p,q} = C^p(X; \pi_p(Y_{p+q}))$. It is easy to see that d_1 is just the ordinary co-boundary; it follows that $E_2^{p,q} = H^p(X; \pi_p(Y_{p+q}))$.

4.2. *Proof of Theorem 3.2.* Define a second exact couple $(\bar{A}, \bar{C})$ as follows:

$$\text{let } \bar{C}_{p,q} = [X, Y_{p+q}(p)],$$

$$\bar{A}_{p,q} = [X, Y_{p+q}(0, p)],$$

where the maps

$$\bar{f}_{p,q}: \bar{A}_{p,q} \rightarrow \bar{A}_{p-1, q+1},$$

$$\bar{g}_{p,q}: \bar{A}_{p,q} \rightarrow \bar{C}_{p+2, q-1},$$

and

$$\bar{h}_{p,q}: \bar{C}_{p,q} \rightarrow \bar{A}_{p,q}$$

are induced by the standard maps (defined in § 2.2)

$$Y_{p+q}(0, p) \rightarrow Y_{p+q}(0, p-1),$$

$$Y_{p+q}(0, p) \xrightarrow{\delta} Y_{p+q+1}(p+2),$$

and

$$Y_{p+q}(p) \rightarrow Y_{p+q}(0, p),$$

respectively. For convenience, we take $\bar{C}$ to be the E_2 -term of the associated spectral sequence; we obviously have

$$\bar{E}_2^{p,q} = \bar{C}_{p,q} = [X, Y_{p+q}(p)] = H^p(X; \pi_p(Y_{p+q})).$$

4.3. *Proof of Theorem 3.3.* This theorem will follow from

LEMMA 4.3.1. *There exists a couple map $\psi: (A', C') \rightarrow (\bar{A}, \bar{C})$, where (A', C') is the first derived of (A, C) , which preserves both gradings, and is an isomorphism from C' to $\bar{C}$.*

Proof. By definition $A'_{p,q} = \text{Im}[i^*: \bar{H}^{p+q}(X^{p+1}) \rightarrow \bar{H}^{p+q}(X^p)]$. Consider the commutative diagram

$$\begin{array}{ccc} [X^{p+1}, Y_{p+q}] & \xrightarrow{i^*} & [X^p, Y_{p+q}] \\ \theta \downarrow & & \theta \downarrow \\ [X^{p+1}, Y_{p+q}(0, p)] & \xrightarrow{\bar{i}} & [X^p, Y_{p+q}(0, p)], \end{array}$$

where i^* and $\bar{i}$ are induced by the inclusion $i: X^p \rightarrow X^{p+1}$, and θ is induced by the standard map $Y_{p+q}(0, \infty) \rightarrow Y_{p+q}(0, p)$. Thus $(\text{Im } i^*) \theta \subset \text{Im } \bar{i}$. But $\bar{i}$ is a monomorphism, since its kernel is

$$\text{Im} [[X^{p+1}/X^p, Y_{p+q}(0, p)]_0 \rightarrow [X^{p+1}, Y_{p+q}(0, p)]],$$

and

$$\begin{aligned} [X^{p+1}/X^p, Y_{p+q}(0, p)]_0 &= C^{p+1}(X; \pi_{p+1}(Y_{p+q}(0, p))) \\ &= 0. \end{aligned}$$

Thus θ induces a map $\bar{\theta}: A'_{p,q} \rightarrow [X^{p+1}, Y_{p+q}(0, p)]$. But

$$[X^{p+1}, Y_{p+q}(0, p)] \cong [X, Y_{p+q}(0, p)],$$

since any map $X^{p+1} \rightarrow Y_{p+q}(0, p)$ can be extended to a map $X \rightarrow Y_{p+q}(0, p)$, and this extension is unique up to homotopy. It follows that θ induces a map $\psi_A: A'_{p,q} \rightarrow \bar{A}_{p,q}$; we define the map $\psi_C: C'_{p,q} \rightarrow \bar{C}_{p,q}$ to be the identity. It remains to prove that ψ is a couple map, i.e. we have the 3 properties

- (a) $f'_{p,q} \psi_A = \psi_A \bar{f}_{p,q}$,
- (b) $g'_{p,q} \psi_C = \psi_C \bar{g}_{p,q}$,
- (c) $h'_{p,q} \psi_A = \psi_C \bar{h}_{p,q}$,

where f', g' , and h' are the maps derived from f, g and h , respectively (see (9)).

Proof of (a).

Consider the commutative diagram

$$\begin{array}{ccccccc} [X^{p+1}, Y_{p+q}] & \xrightarrow{i_1^*} & [X^p, Y_{p+q}] & \xrightarrow{i^*} & [X^{p-1}, Y_{p+q}] & & \\ \theta \swarrow & & \theta \swarrow & & \theta \swarrow & & \\ [X^{p+1}, Y_{p+q}(0, p)] & \xrightarrow{\bar{i}_1} & [X^p, Y_{p+q}(0, p)] & \longrightarrow & [X^p, Y_{p+q}(0, p-1)] & \xrightarrow{\bar{i}_2} & [X^{p-1}, Y_{p+q}(0, p-1)] \\ \cong \swarrow & & \nwarrow & & \nwarrow & & \\ & [X, Y_{p+q}(0, p)] & \xrightarrow{\bar{f}_{p,q}} & [X, Y_{p+q}(0, p-1)] & & & \end{array}$$

Here, the maps are induced by the obvious inclusions and fibre maps. We have $A'_{p,q} = \text{Im } i_1^*$, $A'_{p-1,q+1} = \text{Im } i_2^*$, and $f'_{p,q} = i_2^* | (\text{Im } i_1^*)$. Further, $\bar{i}_1$ and $\bar{i}_2$ are monomorphisms. Hence obviously $f'_{p,q} \psi_A = \psi_A \bar{f}_{p,q}$.

Proof of (b).

$g'_{p,q}: A'_{p,q} \rightarrow C'_{p+2,q-1}$ is induced by

$$g_{p+1,q-1}: \bar{H}^{p+q}(X^{p+1}) \rightarrow \bar{H}^{p+q+1}(X^{p+2}, X^{p+1}).$$

It is sufficient to prove the following diagram commutative

$$\begin{array}{ccc} [X^{p+1}, Y_{p+q}] & \xrightarrow{g_{p+1,q-1}} & [X^{p+2}/X^{p+1}, Y_{p+q+1}]_0 \cong [X^{p+2}/X^{p+1}, Y_{p+q+1}(p+2)]_0 \\ \downarrow \theta & & \downarrow j^* \\ [X^{p+1}, Y_{p+q}(0, p)] & \cong [X^{p+2}, Y_{p+q}(0, p)] \rightarrow [X^{p+2}, Y_{p+q+1}(p-2)] \end{array}$$

since j^* is just the natural projection

$$\begin{aligned} [X^{p+2}/X^{p+1}, Y_{p+q+1}(p+2)]_0 &= C^{p+2}(X; \pi_{p+2}(Y_{p+q+1})) \rightarrow \frac{C^{p+2}(X; \pi_{p+2}(Y_{p+q+1}))}{B^{p+2}(X; \pi_{p+2}(Y_{p+q+1}))} \\ &= [X^{p+2}, Y_{p+q+1}(p+2)]. \end{aligned}$$

This obviously reduces to proving the following diagram commutative

$$\begin{array}{ccc} [X^{p+1}, Y_{p+q}(0, p+1)] & \xrightarrow{\delta^*} [X^{p+2}/X^{p+1}, Y_{p+q+1}(0, p+2)]_0 \xleftarrow{\cong} [X^{p+2}/X^{p+1}, Y_{p+q+1}(p+2)]_0 \\ \searrow \bar{\alpha} & & \downarrow j^* \\ [X^{p+1}, Y_{p+q}(0, p)] & \xleftarrow{\cong} [X^{p+2}, Y_{p+q}(0, p)] \xrightarrow{\bar{\delta}} [X^{p+2}, Y_{p+q+1}(p+2)] \end{array}$$

the maps being the obvious ones.

Now an element x of $[X^{p+1}, Y_{p+q}(0, p+1)]$ may be represented either by a map $\mu: X^{p+1} \rightarrow Y_{p+q}(0, p+1)$, or a map $\mu': SX^{p+1} \rightarrow Y_{p+q+1}(0, p+2)$, in virtue of the canonical isomorphism $[A, \Omega B] \cong [SA, B]$. Consider the following diagram, where the right-hand column is an exact sequence of spaces.

$$\begin{array}{ccc} X^{p+1} & \xrightarrow{\mu} & Y_{p+q}(0, p+1) \\ i \downarrow & & \downarrow \alpha \\ X^{p+2} & \xrightarrow{\rho} & Y_{p+q}(0, p) \\ Pi \downarrow & & \downarrow \delta \\ X^{p+2}/X^{p+1} & \xrightarrow{\nu} & Y_{p+q+1}(p+2) \\ Qi \downarrow & & \downarrow \beta \\ SX^{p+1} & \xrightarrow{\mu'} & Y_{p+q+1}(0, p+2) \end{array}$$

The element $(x)\delta^*$ is represented by the map $(Qi)\mu'$, which lifts uniquely to

$$\nu: X^{p+2}/X^{p+1} \rightarrow Y_{p+q+1}(p+2),$$

making $\nu\beta \simeq (Qi)\mu'$. Then $(x)\delta^*j^*$ is represented by $(Pi)\nu$. On the other hand, $\mu\alpha$ extends uniquely to $\rho: X^{p+2} \rightarrow Y_{p+q}(0, p)$, and $\rho\delta \simeq (Pi)\nu$. Thus

$$\begin{aligned} (x)\delta^*j^* &= [(Pi)\nu] \\ &= [\rho\delta] \\ &= (x)\bar{\alpha}\bar{\delta}, \quad \text{which completes the proof of (b).} \end{aligned}$$

Proof of (c).

$$h'_{p,q}: C'_{p,q} \rightarrow A'_{p,q} = \text{Im} [i^*: \bar{H}^{p+q}(X^{p+1}) \rightarrow \bar{H}^{p+q}(X^p)]$$

is induced by $h_{p,q}: \bar{H}^{p+q}(X^p, X^{p-1}) \rightarrow \bar{H}^{p+q}(X^p)$. Consider the following commutative diagram, where both rows are exact.

$$\begin{array}{ccccc} [X^{p+1}, Y_{p+q}] & \xrightarrow{i^*} & [X^p, Y_{p+q}] & \xrightarrow{\delta^*} & [X^{p+1}/X^p, Y_{p+q+1}]_0 \\ \downarrow \theta & & \downarrow \theta & \nearrow h_{p,q} & \downarrow \theta \\ & & [X^p/X^{p-1}, Y_{p+q}]_0 & \xrightarrow{\cong} & [X^p/X^{p-1}, Y_{p+q}(0, p)]_0 \\ & & \searrow \bar{j} & & \\ 0 \longrightarrow [X^{p+1}, Y_{p+q}(0, p)] & \xrightarrow{\bar{i}} & [X^p, Y_{p+q}(0, p)] & \xrightarrow{\bar{\delta}} & [X^{p+1}/X^p, Y_{p+q+1}(0, p+1)]_0 \end{array}$$

Here, both $\bar{j}\bar{\delta}$ and $h_{p,q}\delta^*$ are the ordinary co-boundary d . Thus $h'_{p,q}\psi_A$ is induced by $h_{p,q}\theta(\bar{i})^{-1} = \bar{j}(\bar{\delta})^{-1}$, which in turn clearly induces the standard map

$$\bar{h}_{p,q}: [X, Y_{p+q}(p)] \rightarrow [X, Y_{p+q}(0, p)].$$

It follows that $h'_{p,q}\psi_A = \psi_C \bar{h}_{p,q}$, which completes the proof of Lemma 4.3.1, and hence of Theorem 3.3.

4.4. *Proof of Theorem 3.4.* An alternative description of $\bar{E}_r^{p,q}$ is $\bar{E}_r^{p,q} = \bar{Z}_r^{p,q}/\bar{B}_r^{p,q}$, where

$$\begin{aligned} \bar{Z}_r^{p,q} &= \text{Im} [[X, Y_{p+q}(p, p+r+2)] \rightarrow [X, Y_{p+q}(p)]] \\ \bar{B}_r^{p,q} &= \text{Im} [[X, Y_{p+q-1}(p-r+1, p-2)] \rightarrow [X, Y_{p+q}(p)]] \\ &= \text{Ker} [[X, Y_{p+q}(p)] \rightarrow [X, Y_{p+q}(p-r+2, p)]] \end{aligned}$$

$\bar{d}_r: \bar{E}_r^{p,q} \rightarrow \bar{E}_r^{p+q, q-r+1}$ is then defined by the commutative diagram

$$\begin{array}{ccccc} [X, Y_{p+q}(p, p+r+2)] & \xrightarrow{\eta_1} & [X, Y_{p+q}(p)] & \xrightarrow{\zeta_1} & [X, Y_{p+q}(p-r+2, p)] \\ \downarrow \delta_1 & & & & \downarrow \delta_2 \\ [X, Y_{p+q+1}(p+r, p+2r-2)] & \xrightarrow{\eta_2} & [X, Y_{p+q+1}(p+r)] & \xrightarrow{\zeta_2} & [X, Y_{p+q+1}(p+2, p+r)] \end{array}$$

Here,
$$\frac{\text{Im } \eta_1}{\text{Ker } \zeta_1} \cong \text{Im } (\eta_1 \zeta_1) \xrightarrow{\delta_3} \text{Im } (\eta_2 \zeta_2) \cong \frac{\text{Im } \eta_2}{\text{Ker } \zeta_2},$$

the composite being $\bar{d}_r$.

Since the diagram is commutative, $\bar{d}_r$ is also the result of chasing round the diagram in the opposite direction; to be precise, let x be an element of $\text{Im } \eta_1/\text{Ker } \zeta_1$. Lift this up to $y \in \text{Im } \eta_1$, and then to $z \in [X, Y_{p+q}(p, p+r+2)]$; then $(x)\bar{d}_r = (z)\delta_1\eta_2\lambda$, where λ is the projection $\text{Im } \eta_2 \rightarrow \text{Im } \eta_2/\text{Ker } \zeta_2$, and this is independent of the choice involved in picking y and z . But $\delta_1\eta_2 = k^{p, p+r-2}$, the k -invariant in

$$H^{p+r}(Y_{p+q}(p, p+r-2); \pi_{p+r-1}(Y_{p+q})).$$

Thus $(x)\bar{d}_r = ([k^{p, p+r-2}]y)\lambda$, which proves Theorem 3.4.

5. *Examples.* 5.1. *Cohomotopy groups.* The spectral sequence $H^*(X) \Rightarrow \pi^*(X)$ has already been studied by Massey (9) and Peterson (10). Restricting ourselves to a suitable range of dimensions, the spectral sequence is given by Theorem 3.1, by taking

$$Y_{p+q} = S^{p+q} \quad (q \leq 0).$$

We can immediately identify $d_2: E_2^{p,0} \rightarrow E_2^{p+2,-1}$ with $Sq^2: H^p(X; Z) \rightarrow H^{p+2}(X; Z_2)$, and $d_3: E_3^{p,0} \rightarrow E_3^{p+3,-2}$ with the Adem operation (see (2))

$$\Phi: \text{Ker}[Sq^2: H^p(X; Z) \rightarrow H^{p+2}(X; Z_2)] \rightarrow H^{p+3}(X; Z_2).$$

5.2. $K_c^*(X)$. According to Adams (1), the spaces $\Omega^{2q}BU(2q+2, \infty)$, $BU(2, \infty) = BU$ are equivalent, and the k -invariant $k^{2q+2r+1} \in H^{2q+2r+1}(BU(2q+2, 2q+2r-2); Z)$ determines the k -invariant $\sigma^{2q}k^{2q+2r+1} \in H^{2r+1}(\Omega^{2q}BU(2q+2, 2q+2r-2); Z)$ and hence the k -invariant $k^{2r+1} \in H^{2r+1}(BU(2, 2r-2); Z)$. Thus all the differentials in the spectral sequence $H^*(X) \Rightarrow K^*(X)$ are *stable* operations, given by the k -invariants of $BU(2q+2, \infty)$, where q is large. We may then use Adams's result in (1) to reproduce the information in (4); in particular $d_3 = Sq^3$ (d_r vanishes for r even, since $\bar{E}_r^{p,q} = 0$ for q odd).

Further, we can prove that the differentials are *derivations* (this has been proved by Dold (6), and, he reports, by Atiyah): for the multiplication in $K^*(X)$ is given by the H -space structure of BU , and according to Kahn (8), all k -invariants of an H -space are primitive.

REFERENCES

- (1) ADAMS, J. F. On Chern characters and the structure of the unitary group. *Proc. Cambridge Philos. Soc.* **57** (1961), 189–199.
- (2) ADEM, J. The iteration of Steenrod squares in algebraic topology. *Proc. Nat. Acad. Sci. U.S.A.* **38** (1952), 720–726.
- (3) ATIYAH, M. F. and HIRZEBRUCH, F. Vector bundles and homogeneous spaces. *Proceedings of Symposia in Pure Mathematics*, vol. III (American Mathematical Society, Providence, R.I., 1960), pp. 7–38.
- (4) ATIYAH, M. F. and HIRZEBRUCH, F. Analytic cycles on complex manifolds. *Topology*, **1** (1962), 25–45.
- (5) BROWN, E. H. Cohomology theories. *Ann. of Math.* **75** (1962), 467–484.
- (6) DOLD, A. Relations between ordinary and extraordinary homology. *Colloquium on Algebraic Topology, Aarhus*, 1962 (mimeographed notes), pp. 2–9.
- (7) EILENBERG, S. and STEENROD, N. *Foundations of algebraic topology* (Princeton, 1952).
- (8) KAHN, D. W. Induced maps for Postnikov systems. *Colloquium on Algebraic Topology, Aarhus*, 1962 (mimeographed notes), pp. 47–51.
- (9) MASSEY, W. S. Exact couples in algebraic topology. *Ann. of Math.* **56** (1952), 363–396; **57** (1953), 248–286.
- (10) PETERSON, F. P. Some results on cohomotopy groups. *American J. Math.* **78** (1956), 243–258.
- (11) PETERSON, F. P. Functional cohomology operations. *Trans. American Math. Soc.* **86** (1957), 197–211.
- (12) PETERSON, F. P. and STEIN, N. Secondary cohomology operations: two formulas. *American J. Math.* **81** (1959), 281–305.
- (13) PUPPE, D. Homotopiemengen und ihre induzierten Abbildungen. I. *Math. Z.* **69** (1958), 199–344.
- (14) WHITEHEAD, J. H. C. Combinatorial homotopy. I. *Bull. American Math. Soc.* **55** (1949), 213–245.